

Technical Document #336: Software Engineering - Comprehensive Overview

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Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Microservices architecture structures applications as independently deployable services.
- Code review examines code changes before merging.
- Continuous integration (CI) automatically builds and tests code changes.